

MANAKSETU

Technical Approach & Technical Documentation

Digital Verification & Certification Platform for Weighing & Measuring Instruments

Prepared from the current ManakSetu project materials: the previous technical document and the supplied Smart India Hackathon 2026 presentation.

1. Purpose & Solution

ManakSetu is a unified digital platform designed to connect citizens/customers, manufacturers, vendors/shopkeepers, Legal Metrology Officers (LMOs), Government Approved Test Centres (GATCs), and administrators. It digitizes the journey from instrument registration and verification requests to field verification, digital certification and public QR-based status checking. The platform is intended to make verification more traceable, transparent and easier to monitor.

2. Platform Architecture

The system is organized as a role-based web platform with separate experiences for each stakeholder and a centralized data and workflow layer.

Layer	Technology / Approach	Main Role
User Interface	React.js, Tailwind CSS, React Router, Axios, responsive design	Role-based portals, forms, dashboards, search and status tracking
Application Layer	Node.js, Express.js, REST APIs	Authentication, authorization, instrument registration, requests, scheduling, verification and reporting
Data Layer	PostgreSQL / centralized database	Users, instruments, applications, verification records, certificates, status history and reports
Verification	Unique certificate ID, QR code, signed/hash-based integrity	Public verification and protection against altered or duplicate records
Deployment	Cloud deployment such as Render/Railway	Scalable hosting, backups and monitored services

3. Stakeholder Portals & Responsibilities

Stakeholder	Key Functions
Citizen / Customer	Search or scan QR, view instrument status, submit complaints and track relevant requests.
Manufacturer	Pre-register instruments and associate registration/QR information at source.
Vendor / Shopkeeper	Register instruments, submit verification requests, request visits and receive expiry reminders.
LMO	Review assigned work, visit and inspect instruments, record results digitally and manage field verification.
GATC	Support verification-only operations and centralized slot/appointment handling where applicable.
Administrator / Government	Monitor applications, pendency, verification activity, records, reports and overall system performance.

4. End-to-End Operational Workflow

The core journey presented in the project deck is a single connected process from registration/application to verification and public trust.

Step	Digital Process
1	Instrument registration / pre-registration
2	Verification request or complaint submission
3	Authority review and request prioritization / scheduling
4	Officer or authorized centre performs verification and records findings
5	Verification result is stored and status is updated
6	Successful verification produces a digital certificate with a unique QR
7	Citizen scans QR to view the authoritative current status
8	Expiry reminders, follow-up checks and administrative monitoring support continued compliance

5. Complaint, Follow-up & Status Monitoring

ManakSetu includes a direct complaint channel for suspected issues. A complaint can move through authority review and, where verification is required, into the officer workflow. Application, verification and certificate status should remain traceable so stakeholders can see the current stage and administrators can monitor pending work.

6. QR & Digital Certificate

Each verified instrument can be associated with a unique digital certificate and QR code. The QR should open a public verification endpoint that retrieves the authoritative record and displays the current instrument status. A signed or hash-based integrity mechanism can be used to help detect unauthorized modification or duplicate records.

7. Security & Operational Reliability

Control	Approach
Access control	Role-Based Access Control (RBAC); each stakeholder receives only required permissions.
Data protection	HTTPS/TLS in transit, strong password hashing and protected environment secrets.
API security	Server-side validation, authorization checks, controlled CORS, rate limiting where appropriate and protection against common abuse.
Certificate integrity	Unique certificate identifiers with QR verification and hash/signature validation.
Auditability	Important verification decisions, certificate issuance, status changes and administrative actions are traceable.
Files & database	Validate uploaded files, use safe database access patterns, least-privilege credentials and backups.

8. Deployment, Scalability & Field Operations

The project materials describe a cloud-first deployment model with separate development, testing and production environments. Version control, validation/testing, monitored services, managed database backups and controlled configuration support reliable deployment.

For field work in low-connectivity areas, the officer interface is designed around an offline-first approach: permitted actions can be queued on the device and synchronized when connectivity returns, with timestamps, conflict handling and server-side validation.

9. Key Benefits

- Transparent verification records and easier public trust.
- End-to-end tracking from request to certificate.
- Reduced manual coordination and missed follow-ups.
- Faster public status checking through QR verification.
- Better administrative visibility through dashboards, reports and pendency monitoring.
- Support for low-connectivity field work through an offline-first synchronization approach.

10. Technical Stack

Area	Technologies / Approach
Frontend	React.js • Tailwind CSS • React Router • Axios • Responsive UI
Backend	Node.js • Express.js • REST APIs • File handling • Report generation
Database	PostgreSQL • Centralized data storage
Verification	QR Code • Unique certificate ID • Signed/hash-based integrity
Deployment	Cloud hosting such as Render/Railway • Backups • Scalable infrastructure

11. Expected Outcome

ManakSetu converts a fragmented verification journey into a connected digital workflow with traceable records, role-based operations, digital certificates, QR-based public verification, complaint handling and administrative oversight. The design is intended to improve transparency, timely verification, citizen trust and data-driven monitoring.

Source alignment note: The stakeholder roles, workflow, technology stack, QR approach, offline field concept, feasibility considerations and benefits in this document are based on the supplied ManakSetu project presentation and the earlier technical documentation. The live Vercel landing page was also checked, but its internal application screens were not exposed as readable page content during review; therefore no unverified internal website feature has been presented as a confirmed implementation.